Experiment Number: 2 MATLAB Simulation of Cellular Coverage, Cell Splitting and Handoff in a Multi-Cell Network

PROGRAM:

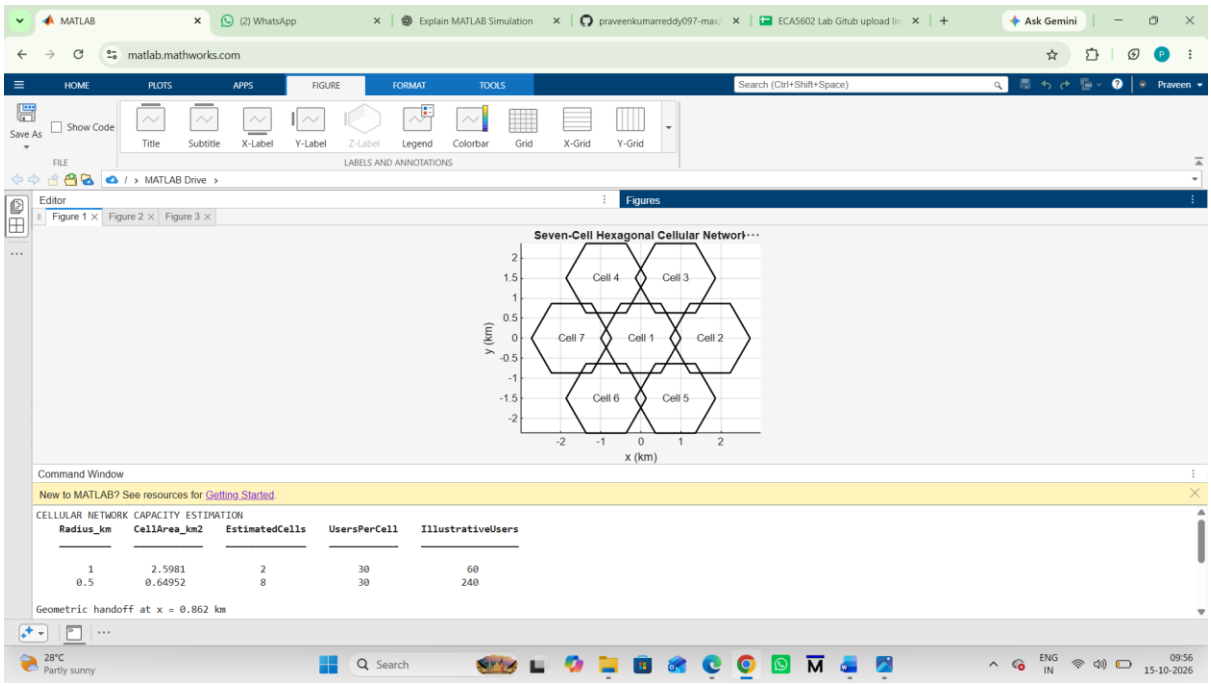
```
% Experiment 2: Simplified Hexagonal Coverage and Handoff
clear;
clc;
close all;
%% 1. Baseline assumptions
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
% Area of a regular hexagonal cell
hexArea = @(R) (3*sqrt(3)/2) .* R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
% Display capacity table
capacityTable = table( ...
    radiiKm(:), ...
    cellAreaKm2(:), ...
    cellCount(:), ...
    repmat(usersPerCell, numel(radiiKm), 1), ...
    estimatedUsers(:), ...
    'VariableNames', ...
    {'Radius_km', 'CellArea_km2', 'EstimatedCells', ...
     'UsersPerCell', 'IllustrativeUsers'});

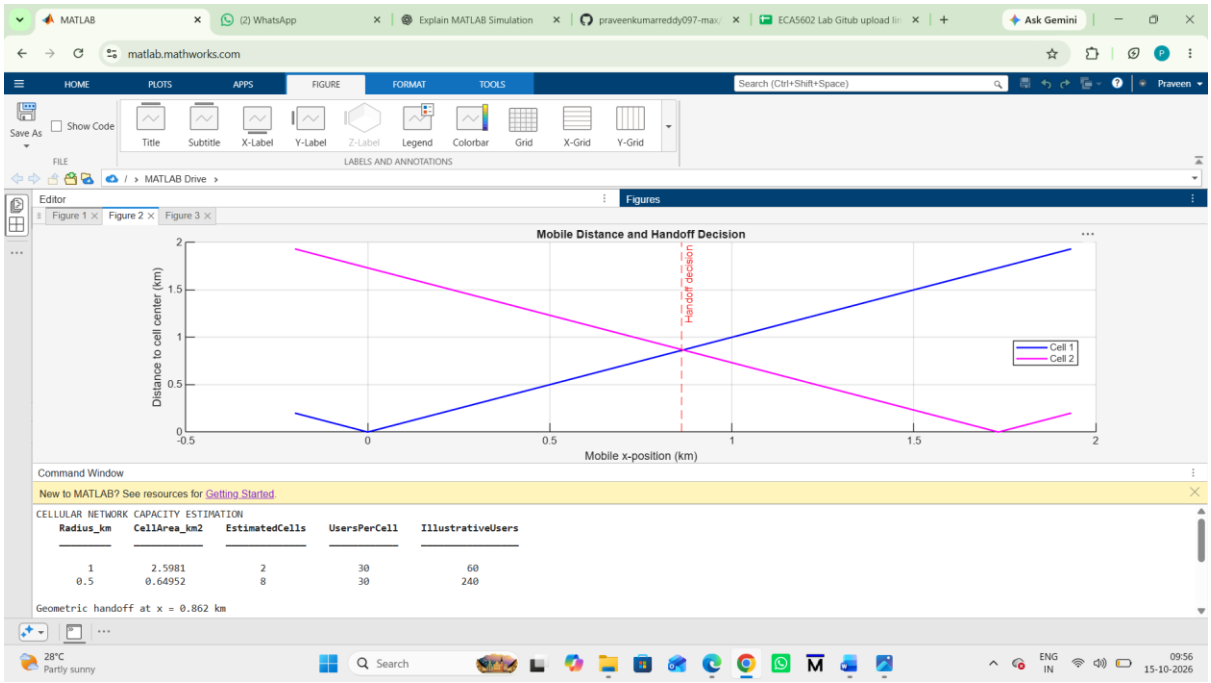
disp('CELLULAR COVERAGE AND CAPACITY');
disp(capacityTable);
%% 2. Seven-cell hexagonal cluster
R = radiiKm(1);
% Flat-top hexagons:
% One central cell and six surrounding cells
centers = R * [ ...
     0,          0;
     sqrt(3),    0;
     sqrt(3)/2,  1.5;
    -sqrt(3)/2,  1.5;
     sqrt(3)/2, -1.5;
    -sqrt(3)/2, -1.5;
    -sqrt(3),    0];
```

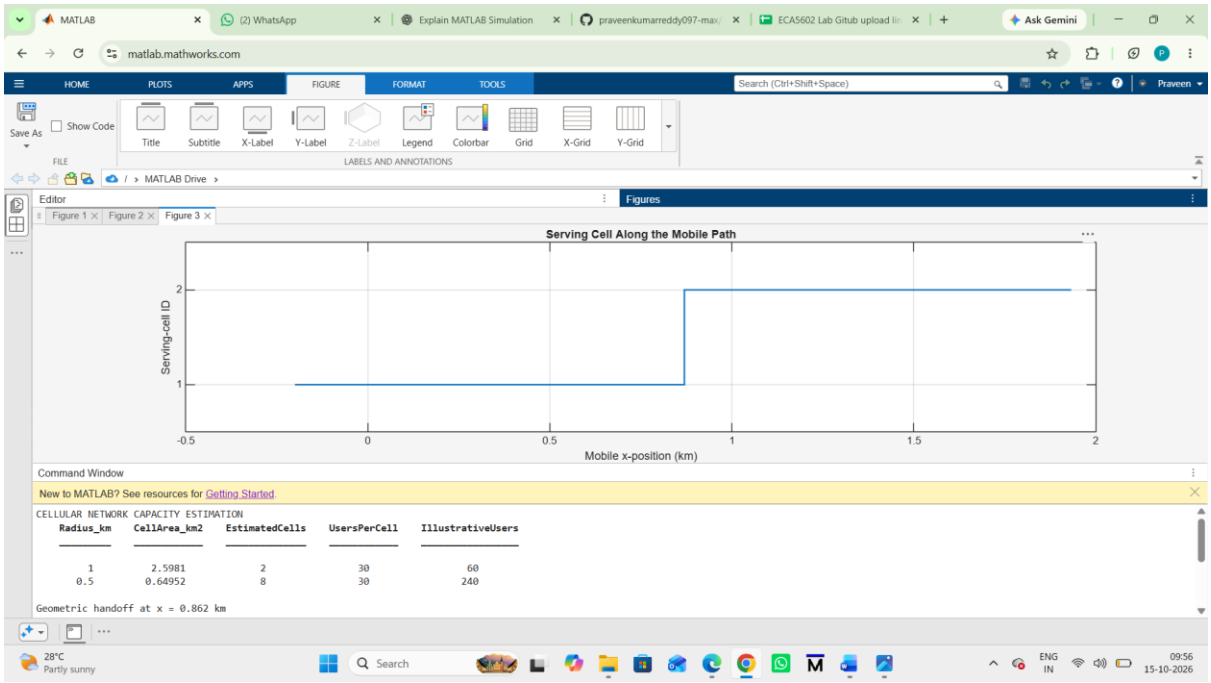
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theta = (0:6) * pi/3;
figure;
hold on;
axis equal;
grid on;
for k = 1:size(centers, 1)
    xHex = centers(k,1) + R*cos(theta);
    yHex = centers(k,2) + R*sin(theta);
    plot(xHex, yHex, 'k-', 'LineWidth', 1.5);
    text(centers(k,1), centers(k,2), ...
        sprintf('Cell %d', k), ...
        'HorizontalAlignment', 'center', ...
        'VerticalAlignment', 'middle');
end
xlabel('x (km)');
ylabel('y (km)');
title('Idealized Seven-Cell Hexagonal Cluster');
hold off;
%% 3. Mobile movement between two neighboring cells
cell1 = centers(1,:);
cell2 = centers(2,:);
% Mobile moves along the line joining the cell centers
mobileX = linspace(-0.2*R, sqrt(3)*R + 0.2*R, 300);
mobileY = zeros(size(mobileX));
% Distances from mobile to cell centers
d1 = hypot(mobileX - cell1(1), mobileY - cell1(2));
d2 = hypot(mobileX - cell2(1), mobileY - cell2(2));
% Assign the nearest cell
servingCell = ones(size(mobileX));
servingCell(d2 < d1) = 2;
% Detect the serving-cell transition
handoffIndex = find(diff(servingCell) ~= 0, 1, 'first');
%% 4. Plot distances and handoff decision
figure;
hold on;
grid on;
plot(mobileX, d1, 'b-', 'LineWidth', 1.5);
plot(mobileX, d2, 'm-', 'LineWidth', 1.5);
if ~isempty(handoffIndex)
    xline(mobileX(handoffIndex), '--r', 'Handoff decision');
end
xlabel('Mobile x-position (km)');
ylabel('Distance to cell center (km)');
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legend('Cell 1', 'Cell 2', 'Location', 'best');
title('Nearest-Center Handoff Approximation');
hold off;
%% 5. Plot serving-cell ID along the path
figure;
stairs(mobileX, servingCell, 'LineWidth', 1.5);
grid on;
xlabel('Mobile x-position (km)');
ylabel('Serving-cell ID');
yticks([1 2]);
ylim([0.5 2.5]);
title('Serving Cell Along the Mobile Path');
%% 6. Display handoff position
if ~isempty(handoffIndex)
    fprintf('Geometric handoff at x = %.3f km\n', ...
        mobileX(handoffIndex));
else
    fprintf('No handoff detected.\n');
end
```

OUTPUT:




Mobile Distance and Handoff Decision
Distance to cell center (km)
Mobile x-position (km)
Handoff decision
Cell 1
Cell 2
CELLULAR NETWORK CAPACITY ESTIMATION
Radius_km CellArea_km2 EstimatedCells UsersPerCell IllustrativeUsers
1 2.5981 2 30 60
0.5 0.64952 8 30 240
Geometric handoff at x = 0.862 km


Serving Cell Along the Mobile Path
Serving-cell ID
Mobile x-position (km)
CELLULAR NETWORK CAPACITY ESTIMATION
Radius_km CellArea_km2 EstimatedCells UsersPerCell IllustrativeUsers
1 2.5981 2 30 60
0.5 0.64952 8 30 240
Geometric handoff at x = 0.862 km